A QoE-aware Mechanism to Improve the Dissemination of Live Videos over VANETs

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Abstract. *Real-time video dissemination over Vehicular Ad-hoc NETworks (VANETs) is challenging due to its strict quality requirements, dynamic network topology, and broadcast environment. Although Statistical Routing Protocols (SRPs) have been applied by using positioning and Quality of Service (QoS) parameters, they are not enough to obtain satisfactory Quality of Experience (QoE) levels in multi-hop video dissemination due to different requirements of multimedia services. This paper introduces the QOe-aware REceiver-based (QORE) mechanism to improve the dissemination of real-time videos in VANETs. QORE employs a QoE-driven Unequal Error Protection (UEP) scheme, namely Interleaving, which mitigates the effects of frame loss by spreading out the bursty losses. Further, QORE can be integrated with SRPs to offer QoE-driven parameters for the relay node selection and backbone maintenance. Thus, nodes decide for themselves to retransmit further the video sequences, enhancing the capacity of the system in delivering QoE-aware live videos. QORE was added to a straightforward Distance-based SRP, named DQORE. Results show the gains of DQORE in relation to SRPs, achieving video dissemination with QoE support, less routing overhead, and robustness in V2V VANETs.*

1. Introduction

Nowadays, people are increasingly spending more and more time in their vehicles. This has encouraged industries and manufacturers to provide drivers and passengers with a wide scope of novel real-time multimedia services, ranging from safety and security traffic warnings to live entertainment and advertising videos [Kakkasageri and Manvi 2014]. In this way, efficient on-road live video dissemination support in Vehicular Ad-hoc NETworks (VANETs) has become a trend and, at the same time, necessary for the future of the Intelligent Transportation System (ITS) [Gerla et al. 2014].

To meet this demand, vehicles have employed flooding approaches extensively as data delivery mechanism. Although, the simple flooding of data within the network leads to an explosive growth of traffic resulting in collisions and congestions, i.e., the broadcast storm problem [Torres et al. 2014]. In addition, due to the high vehicle mobility, the delivery of real-time video content, while maximizing the user's Quality of Experience (QoE), becomes an intricate task and has not been taken into account [Felice et al. 2014]. While Quality of Service (QoS) focuses only on packet-based management and delivery statistics, QoE levels of the received videos must be measured by the subjective acceptability of the users, the latter being key to the safety broadcast of traffic warning videos.

The design of a reliable and robust dissemination approach for real-time video dissemination over VANETs with QoE support is not a straightforward task [Quadros et al. 2014]. Due to ad-hoc nature and the highly dynamic topology, when very close neighbor vehicles decide to transmit the same packets, they can interfere with each other and lead to an unnecessary growth of traffic of data, causing different impacts on the video quality and wasting network resources. Thus, the multi-hop routing service must be aware of QoE requirements and network conditions to recover or maintain video quality with a low overhead and with high reachability for the nodes in the flooding area.

On the other hand, most of the current routing protocols use information exchanged by neighbor vehicles to make routing decisions. However, end-to-end routes suffer from frequent interruptions, and the currently available local topology is not always accurate [Slavik et al. 2014]. Statistical Routing Protocols (SRPs) allow improvements on network performance by making a distributed hop-by-hop routing decision [Torres et al. 2014]. In SRPs the selection of relay nodes is carried out by receivers (receiver-based), without requiring static end-to-end routes, and allowing transmissions even in case of topology changes. However, the existing SRPs do not efficiently explore vehicle positioning, network, and QoE-driven parameters. Thus, a QoE-aware mechanism, that employs information such as hierarchy of frames [Yao et al. 2014] and estimation of video distortion (caused by probability of occurrence of loss and loss burstiness) [Tao et al. 2008] can be coupled as a QoE-pointer to support real-time monitoring and forwarding decision at the routing level [Aguiar et al. 2014].

QoE-driven Unequal Error Protection (UEP) schemes can further enhance the advantages of SRPs[Claypool and Zhu 2003, Li et al. 2014], since they use information from the application-layer (video characteristics and requirements) to minimize the effects of packet loss and achieve better video distribution. These schemes allow multimedia dissemination with QoE support even in the presence of dynamic topologies [Rosário et al. 2014, Immich et al. 2014]. This way, the Interleaving UEP scheme improves the QoE levels of live video flows by merging frames of the original video sequence without increasing packet redundancy, thus avoiding the broadcast storm problem. Therefore, a straightforward SRP coupled with a QoE-aware mechanism and an appropriate QoE-driven UEP scheme (Interleaving) in a unified approach, can contribute to video dissemination with better QoE assurance in dynamic topologies.

This paper proposes a QoE-aware mechanism to support the delivery of real-time videos over V2V VANETs, named QOe-aware REceiver-based (QORE) mechanism. It combines application parameters (e.g., different frame importance, frame position, and video distortion estimation), positioning information, and the QoE-driven Interleaving scheme to establish trade-offs between quality and required hops. QORE performs a self-organized use of relay nodes and can be easily integrated with SRPs, maintaining the packet delivery ratio, reacting well to dynamic environments and enhancing or at least maintaining the QoE level of the disseminated videos when compared to non-QoE-driven schemes. QORE was added to a straightforward Distance-based SRP and was evaluated.

The remainder of this paper is organized as follows. Section 2 presents the related work. Section 3 introduces the QORE mechanism and the coupling of QORE with the Distance-based SRP (called DQORE). Simulation setup and results comparing pure SRPs and DQORE are presented in Section 4. Lastly, conclusions are summarized in Section 5.

2. Related Work

There are three fundamental methods that fit into the category of SRPs (i.e., the next relay nodes are chosen through a distributed backoff phase, by comparing a locally measured value with a threshold value): Counter-Based (CB), Distance-Based (DB), and Location-based (LB) methods [Torres et al. 2014]. The DB method uses the distance to the farthest 1-hop neighbor from whom the packets has been sent as a proxy for rebroadcasting. The LB method, in turn, employs sharing of positional information to allow retransmissions in the uncovered area. Lastly, in the CB method, nodes simply count the number of times that each packet is received during the backoff to count the number of neighbors that so far have retransmitted the packets. These methods have been used to design many protocols including some using hybridization with topological approaches.

In [Mohammed et al. 2009], authors published improved versions of the above statistical methods by adding parameters related to density to further reduce the broadcast storm problem. In [Slavik and Mahgoub 2013] authors proposed the Distribution-Adaptive Distance with Channel Quality (DADCQ) protocol. DADCQ is a DB SRP that is adaptive to distribution pattern and channel quality for multihop V2V broadcast. However, all of these approaches do not take into account video-awareness to the video dissemination, i.e., they do not ensure QoE in the received video streaming. Furthermore, it is important to evaluate new video-based routing schemes based on QoE metrics.

In [Rosário et al. 2014], authors proposed the Link quality and Geographical Opportunistic Routing (LINGO) protocol, a receiver-based protocol that uses a multi-criteria scheme to compute the backoff timer and employs a UEP scheme, namely Forward Error Correction (FEC). Similarly, in [Di Felice et al. 2013], authors presented a geocast and contention-based protocol for multimedia dissemination in VANETs, called Dynamic Backbone Assisted (DBA) MAC protocol. In DBA-MAC, the formation of the backbones also uses a multi-criteria approach, such as link quality, vehicles location, and speed. Despite LINGO and DBA-MAC consider the link quality for routing decisions, these protocols do not take into account QoE-based criteria for the relay node selection and backbone maintenance. Further, these approaches use only one sample of packet to define the best forwarder nodes, which can cause false-positive results on the link quality measurement. Although the usage of a UEP scheme in LINGO, the use of FEC may worsen the broadcast storm problem due to increase of redundant packets in the network.

In [Slavik et al. 2014], authors introduce the Distance-to-Mean (DTM) method, which extends the DB method, where nodes favor rebroadcasting when they cover a large amount of physical area that has not been covered by neighboring nodes. Similarly, in [Torres et al. 2014], authors propose Automatic Copies Distance-Based (ACDB), an improved beaconing flooding scheme that extends the CB method to cope with variable vehicle density situations. Besides ACDB uses the Peak signal-to-noise ratio (PSNR) QoE-metric to assess the received videos, a basic limitation of these protocols consists of their reliance on a single criterion to compute the backoff phase, reducing the network reliability for long data transmission, such as live video streaming. Moreover, not only PSNR but also, other real QoE experiments and more sophisticated objective metrics, as shown in Section 4, must be applied to assessment of received videos, since PSNR by itself, does not correlate well with the subjective acceptability of the users [Aguiar et al. 2014]. DTM and ACDB are used as comparison protocols in this work.

From our analysis, a SRP is a promising solution for broadcast in VANETs, since vehicles do not need to flood messages proactively, avoiding broadcast storms. In addition, the existing SRPs do not efficiently combine location information, QoE-awareness, and a UEP scheme for video dissemination. All of these key features are not offered in a unified SRP so far, thus existing proposals lack of robustness and QoE-awareness.

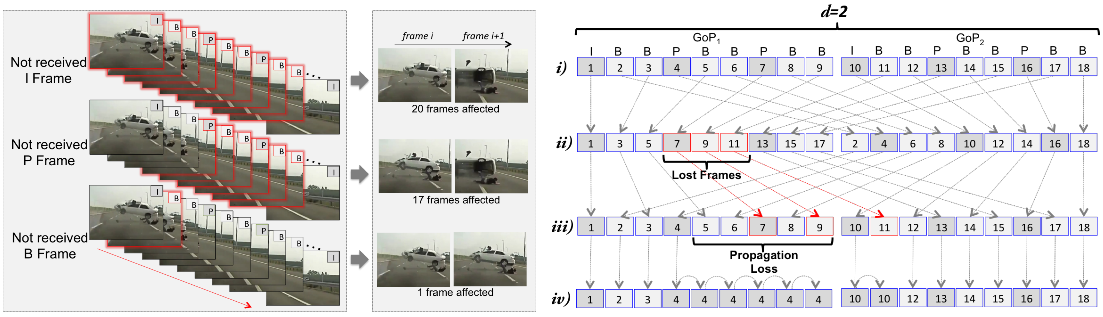
3. The QOe-aware REceiver-based (QORE) mechanism

This section presents the QORE mechanism to improve dissemination of live videos in multi-hop V2V VANETs. It considers a receiver-based approach, where no neighboring information exchange is required. QORE works jointly with an underlying SRP and aims to select relay nodes with high reachability, i.e., nodes that will cover as many destinations in a physical area as possible avoiding unnecessary routing overhead. Further, QORE allows video dissemination with high quality and low impact from the user's point-of-view even in the presence of dynamic topology scenario changes. Depending on the SRP, other different parameters can be incorporated aiming at a more robust decision process. QORE works in three phases, namely QoE-driven Unequal Error Protection (QUEP), Distributed Backoff-based Forwarding (DBF), and Persistent Multi-hop Forwarding (PMF). In the QUEP phase, we employ the video Interleaving method [Claypool and Zhu 2003], where the source node first re-sequences video frames before delivering (to balance the loss of packets) and returns to their original order at the receiver nodes. Upon re-sequencing the video frames in QUEP, in DBF, the source node starts the video flooding and the relay node candidates compete to choose which nodes will participate in the PMF phase. Thus, the chosen nodes retransmit the video sequences further, according to positioning and QoE-driven parameters. The PMF phase provides contention-free forwarding of the video streaming, exploiting the previously built multi-hop paths, and allowing dynamic changes to other paths (relay nodes) in case of link failures and loss of quality. We will detail each phase in the following subsections.

In our network model, let's suppose a scenario of live video dissemination, so that in case of accidents or disasters, vehicles or first responder teams coming toward the crashed area receive videos of accident. Thus, we consider k vehicles containing, each one, an identifier ($i \in [1, k]$), moving over an urban multi-lane grid-area. The combination of those nodes configures a graph of vehicular topology $G(V, E)$, where vertices $V = \{v_1, v_2, ..., v_k\}$ mean a finite subset of k nodes, and edges $E = \{e_1, e_2, ..., e_m\}$ mean a finite set of asymmetric wireless links between them. We denote a subset $N(v_i) \subset V$ as all 1-hop neighbors within the radio range of a given node v_i. Further, each node v_i has an IEEE 802.11p-compliant radio transceiver, through which it can communicate with $N(v_i)$, a GPS (to location awareness and synchronism in time), a multimedia encoder/decoder, and a transmission buffer (*TB*) with a maximum queue capacity (TB_{Max}).

3.1. QoE-driven Unequal Error Protection (QUEP)

Before transmitting, video sequences are compressed by the Motion Picture Expert Group (MPEG) standard in Group of Pictures (GoPs) composed of three frame types, namely I-, P-, and B-frame. Frames between two I-frames define a GoP. I-frames are self-contained, however, to encode and decode P- and B-frames, the previous I-frame and/or P-frames in the same GoP are needed. The compression rate of P-frames is higher than the rate of I-frames, and the compressed rate of B-frames are the highest of all three. Thus, if



(a) (b)

Figure 1. (a) Different priority of frames. (b) Interleaving scheme.

an I- or P-frame is lost, all frames thereafter in the GoP become un-decodable, i.e., the same degree of packet loss may cause severe quality degradation or may pass unnoticed, depending on which frame types are affected. Fig. 1a shows the different degrees of importance for the user's perception in each frame type for a GoP with size 20. I-frames are the most important ones from the human point-of-view. For a single I- or P-frame lost, there is error propagation through frames until the end of the GoP. Though, for a single B frame lost, the impact is not noticed visually, since no others frames are affected.

For the QUEP phase, we consider the Interleaving scheme as QoE-driven UEP mechanism. Because of its modular design, QORE can be easily adapted to use other UEP schemes, since these techniques are carried out before the transmission phase (DBF). We opted for Interleaving due to its low overhead, essential in broadcast environments. This scheme assumes that a better quality from the user's perspective can be achieved by spreading out bursty packet losses in a media flow, i.e., small gaps degrade quality less than a big gap in a multimedia flow. Thus, by using a distance (d), given by the interleaving algorithm, whenever a source node (VS) has a video flow ($VF = \{g_1, g_2, ..., g_n\}$) with n GoPs (g) to send, it first interleaves $d \times |g|$ consecutive frames, i.e., d represents the number of merged GoPs. After that, VS starts the DBF phase, by broadcasting the interleaved frames as flooding. Upon reaching receivers, these frames are then reconstructed back to their original order. If consecutive loss occurs in the interleaved stream during transmission, big gaps can be spread out into several small gaps [Li et al. 2014].

Interleaving is based on the Frame-Copy error concealment technique, where the last well-received frame replaces lost frames. Fig. 1b shows the interleaving mechanism for two ($d = 2$) consecutive GoPs ($|g| = 9$) (***i***)). Upon losing 3 consecutive frames (4th, 5th, and 6th) in the 1st interleaved GoP during the broadcast (***ii***)), the Interleaving scheme breaks the gap into small gaps (***iii***)). Through Frame-Copy, the reconstructed GoPs at receivers are recovered by repeating the previous well-received frame (***iv***)). Otherwise, without interleaving, in case of lost of the 4th, 5th, and 6th frames in ***i***), the gap of lost frames would be only one, but with a much larger size, since the first P-frame (4th frame) is necessary for the reconstruction of the remaining frames in the GoP. Thus, this division of bursty losses in small gaps becomes less noticeable from the user's point-of-view.

3.2. Distributed Backoff-based Forwarding (DBF)

In the DBF phase, the interleaved video flows are disseminated through a contention distributed stage, where VS starts the video transmission and relay nodes (VR) compete among themselves to choose which nodes will participate in the PMF phase. By using

QoE-driven parameters, called QoE-indicators, QORE defines the best VRs in each hop. Once VS begins to capture a given VF, it starts the dissemination in a multi-hop fashion, i.e., VS broadcasts video packets p in a *Time Window*, denoted by $W(VF_i) \subset VF_i$, to all the neighbors of VS ($N(VS)$). The Algorithm 1 presents the generic process of DBF.

Upon receiving a $W(VF_i)$ from a sender node v_a, the nodes $N(v_a)$ have their own and v_a variation of position information, extracted from the packet headers. Thus, a given node $v_b \in N(v_a)$ can easily determine when it is located within the set of nodes in the Forwarding Zone (FZ) of v_a ($FZ(v_a) \subset N(v_a)$) (Line 1 of Algorithm 1), that corresponds to an angle α between the line of movement orientation of v_a and v_b. FZ becomes important, since it limits the selection of VRs to a given sector, avoiding loops and vehicles going to the opposite direction of the crash area. We have defined $\alpha \leq 90^\circ$ to FZ [Wang et al. 2014]. VR candidates have information of the total number of hops traversed N_{Hops} of $W(VF_i)$, i.e., for a $W(VF_i)$ exceeding a predetermined threshold N_{HopsT}, no further rebroadcast occur (Line 1 of Algorithm 1). We have defined $N_{HopsT} = 10$, since for a range of 250m in each hop, we have a medium-long distance of 2.5km approximately, as required in many rescue/disaster VANET scenarios [Felice et al. 2014].

Algorithm 1 DBF phase

When a given node $v_b \in N(v_a)$ receives broadcasted packets ($W(VF_i) = \sum_{k=1}^{n} p_k$) from a node v_a:
1: **if** $v_b \in FZ(v_a)$ **and** $N_{Hops}(W(VF_i)) < N_{HopsT}$ **then**
2:     **if** $\exists!\, p_k \in W(VF_i)$ **and** $p_k \in TB_{v_b}$ **then**
3:         Drop $W(VF_i)$ from TB_{v_b}
4:         **return**
5:     **else**
6:         Compute $FF(v_b)$ (Eq. (6)) **and** Start $BackoffTimer(v_b)$ (Eq. (1))
7:         **while** $BackoffTimer(v_b) \neq 0$ **do**
8:             **if** Overhear ! $p_k \in W(VF_i)$ **then**
9:                 Cancel $BackoffTimer(v_b)$
10:                 **if** $FF > FF_t$ **and** $\angle VR_i v_a v_b \geq \phi$ **then**
11:                     Rebroadcasts $W(VF_i)$ **and** $v_b \leftarrow VR_{i+1}$
12:                 **else**
13:                     Drop $W(VF_i)$ from TB_{v_b} **and** Cancel any new rebroadcast of $p_k \in W(VF_i)$
14:                     **return**
15:                 **end if**
16:             **end if**
17:         **end while**
18:         Rebroadcasts $W(VF_i)$ **and** $v_b \leftarrow VR_i$
19:     **end if**
20: **else**
21:     Drop $W(VF_i)$
22:     **return**
23: **end if**

If $W(VF_i)$ contains only new received packets (Line 2 of Algorithm 1), nodes located within $FZ(v_a)$ apply the *Fit Function* (FF) (Eq. (6)) prior to relay the packets, conversely nodes outside this area drop the received packets from ***TB***. The use of FF allows VR candidates to mitigate the number of retransmissions inside FZ by choosing only the best VRs $\in FZ(v_a)$. The value of FF $[0, FF_{max}]$ depends on parameters of the underlying SRP, such as positioning (Subsection 3.2.2), and QoE-driven parameters, as shown in Subsection 3.2.1. Thus, after calculating the FF, VR candidates replace the location of v_a by their own locations in the packet header, set a *BackoffTimer* according to Eq. (1), and wait for the timeout to rebroadcast the buffered packets ($W(VF_i)$). It is easy to see that nodes with higher values of FF are mapped to smaller *BackoffTimer* sizes, and thus having higher probability to win in the DBF phase.

$$BackoffTimer = CW_{Max} - FF \cdot (CW_{Max} - CW_{Min}) \quad (1)$$

Where CW $[CW_{Min}, CW_{Max}]$ is the size of the *Contention Window* in the 802.11p standard. In a completely distributed manner, the node that generates the smallest *BackoffTimer* rebroadcasts $W(VF_i)$ first and it is selected as VR (line 18 of Algorithm 1). Thus, QORE provides video dissemination via multiple VRs. Moreover, as expected in the IEEE 802.11p standard, VR candidates are able to sense the channel during the *BackoffTimer*, so that they will cancel their countdown timers in case of overhearing transmissions of any $p \in W(VF_i)$ from other nodes. In this case, other nodes in the $FZ(v_a)$ compare their calculated FF to a threshold value (FF_t) and the angle between VR_i, v_a, and v_b to a threshold angle $\phi = 45°$ ($\phi = \alpha/2$), so that these VR candidates can retransmit further the received packets. The FF_t and ϕ values are directly related to the reachability of the underlying SRP. If the calculated FF is bigger than FF_t, and the angle between the previous selected VR and v_b is bigger than ϕ, the VR candidate proceeds to retransmit $W(VF_i)$ itself, otherwise it remains silent.

The QORE mechanism aims to select relay nodes that have better video quality on the user's perspective. Thus, we used a fixed $FF_t = 0.4$ [Slavik et al. 2014]. As outlined in Subsection 3.2.2, to avoid duplicated packets in the same geographic area, the DB method establishes that the farthest nodes (from the previous sender node) rebroadcast further the same $W(VF_i)$. Moreover, through a *passive acknowledgment* approach [Rosário et al. 2014], v_a is also able to overhear the further relaying of $W(VF_i)$ and, thus, concluding that it was successfully received by other nodes in the $FZ(V_a)$, allowing QORE to reduce the acknowledgments in the MAC-layer.

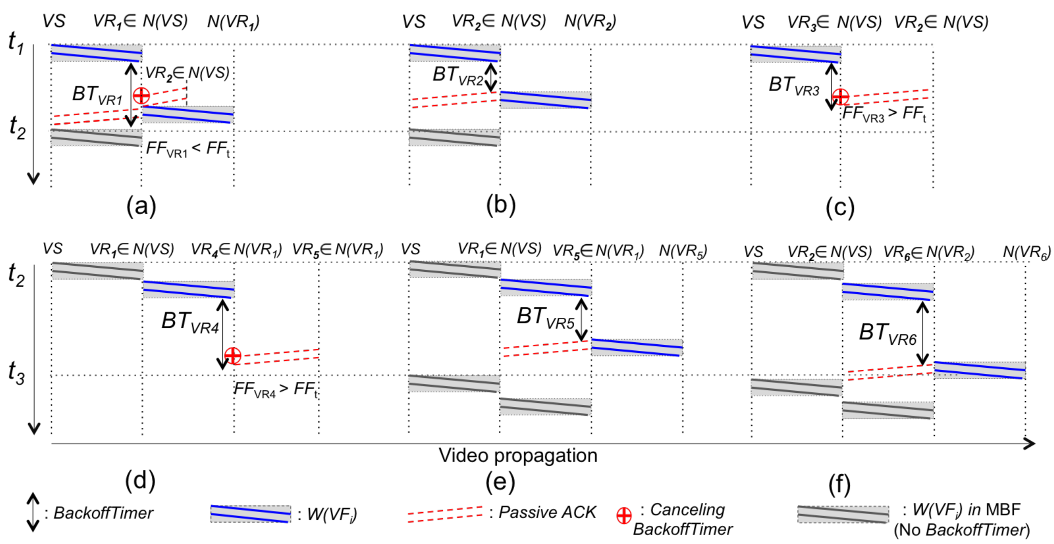


Figure 2. DBF phase for the 1st hop (a-c), and for the 2nd hop (d-f)

Fig. 2 shows an overview of the DBF phase. Suppose that $N(VS) \cap FZ(VS) =$

$\{VR_1, VR_2, VR_3\}$, i.e., nodes VR_1, VR_2, and VR_3 are located in the Forwarding Zone of the source vehicle. In this example, VR_2 forwarded a given $W(VF_i)$ first (t_1 in Fig. 2b) and the neighboring nodes (VR_1 and VR_3) overhear this transmission (t_1 in Figs. 2a and c). As a result, these nodes cancel their $BackoffTimers$ and check out whether their calculated FF are smaller than the FF_t. The neighboring node which has a FF smaller than FF_t, proceed to retransmit $W(VF_i)$ (t_2 in Fig. 2a), otherwise it drops $W(VF_i)$ from its TB and remains silent (line 13 of Algorithm 1 and t_2 in Fig. 2c).

In t_2, for the second hop (Figs. 2d-f), suppose that $N(VR_1) \cap FZ(VR_1) = \{VR_4, VR_5\}$ and $N(VR_2) \cap FZ(VR_2) = \{VR_6\}$, i.e., VR_4 and VR_5 are located in the FZ of VR_1; and only VR_6 is located in the FZ of VR_2. Initially, VR_5 wins the contention phase with the smallest $BackoffTimer$ and forwards $W(VF_i)$ first. Then, VR_1 uses the transmitted packets as *passive acknowledgement*. Meanwhile, the second VR candidate (VR_4, from Fig. 2d) overhears VR_5 transmitting $W(VF_i)$, cancels its $BackoffTimer$ and checks out whether its calculated FF is smaller than the FF_t. As FF_{VR4} is bigger than FF_t, VR_4 remains silent. Finally, in t_3 of Figs. 2d and e, VS transmits subsequent packets to VR_5 and VR_6 by VR_1 and VR_2 without requiring $BackoffTimers$, i.e., it switches to the PMF phase, detailed in Subsection 3.3.

3.2.1. QoE-Indicators

An MPEG video sequence is composed of frames with different degrees of importance for the user's perception, as seen previously. In a 24 fps MPEG-4 video, if packets from one I-frame are lost, the video will be degraded in at least 0.75 s. Also, the loss of P-frames at the beginning of a GoP causes a higher video distortion than loss at the end of a GoP. However, the loss of B-frames does not impact heavily on the user's perception. By considering the importance of each video frame, as well as the P-frame position within the GoP, QORE prioritizes frames with a greater impact on the average video distortion (σ_s^2), as opposed to those with lower QoE impact. Thus, it assigns different weights to slices (s) of packets belonging to each frame as modeled by Eq. (2):

$$\sigma_s^2 \propto \begin{cases} \frac{\alpha_1(R_M - R_I)}{R_M} & \text{if s} \in \text{I-frame} \\ \frac{\alpha_2}{(2^{T-1}-1)R_M} \sum_{i=1}^{T-1} 2^{T-1-i}(R_M - R_{P_i}) & \text{if s} \in \text{P-frame} \\ \frac{\alpha_3(R_M - R_B)}{R_M} & \text{if s} \in \text{B-frame} \end{cases} \tag{2}$$

Where $T-1$ is the number of P-frames per GoP, R_I, R_{P_i}, and R_B mean the I, P (with position i in the GoP), and B-frame received rate in $W(VF_i)$, respectively. R_M is the maximum data-rate supported by the radio transceiver of each vehicle, e.g., for a DCMA-86P2 802.11p Wi-Fi card, $R_M = 6$ Mbps if $P_{rx} >$ -93dbm [Di Felice et al. 2013]. The parameters α_1, α_2, and α_3 are weighting factors, where $\sum_{i=1}^{3} \alpha_i = 1$.

The distortion model proposed by Shu Tao [Tao et al. 2008] considers the impact caused by the loss of single slices of a frame from a video stream. Thus, for a given video frame structure and a probability of occurrence of loss, Eq. (3) defines an overall distortion value for the whole stream, where s and L are the number of slices per video packet and the number of packets per frame, respectively obtained from the standard video codec configurations and packetization. The parameter $\overline{n}$ represents the loss burstiness

($1.06 \geq \overline{n} \geq 1$ for Bernoulli losses, depending on the aggressiveness of the burst errors). The attenuation factor γ ($\gamma < 1$) accounts for the effect of spatial filtering, and varies as a function of the video characteristics and decoder processing. P_e is the probability of loss events (of any length) in the video stream. Both, γ and P_e are given by the effect of the loss pattern experienced by the video stream and the codec's error concealment technique. Finally, the Mean Square Error (MSE) distortion $\overline{D}$ provides a QoE-estimate by using a non-linear relation that measures the video quality level by comparing distortions caused by slice losses [Tao et al. 2008], according to each frame type, denoted in Eq. (4):

$$\overline{D} = sL \cdot \overline{n}P_e \cdot \sigma_s^2 \cdot \left(\frac{\gamma^{-t+1} - (T+1)\gamma + T}{T(1-\gamma)^2} \right) \quad (3)$$

$$C_{QoE}^{v_b}(W(VF_i)) = \frac{1}{1 + exp(b_1 \cdot 10 \cdot log_{10}(255^2/\overline{D}) - b_2)} \quad (4)$$

Where, b_1 is the slope of the QoE mapping curve and b_2 is the central point. By considering 40 dB as the highest video quality, and the lowest video quality for values below 20 dB, the values of b_1 and b_2 are given by 0.5 and 30, respectively. Based on the average distortion caused by losses in the different frame types in $W(VF_i)$ it is possible to VR candidates to compute a higher FF to nodes receiving the most important packets.

3.2.2. Integration of the DBF phase of QORE with the DB SRP

Here we present the QORE mechanism operating together with an underlying routing protocol. To assess QORE functionalities, we develop and adapt its DBF phase jointly with a straightforward broadcast protocol built using the DB method, called DQORE protocol. The DB method calculates coverage through the distance (C_D) from the VR candidate to the previous sender node. When C_D is small, it means the VR candidate is close to the last sender, indicating it should not favor rebroadcasting. Local positional information is used in the DB method to calculate C_D in VR candidates. Thus, the spatial distance is defined in Eq. (5) for a VR candidate positioned at (x, y) and a previous sender node located at $(\overline{x}, \overline{y})$ and normalized to a value between zero and one by dividing by the maximum transmitting range (R). Hence, VR candidates with large geographical distance from previous sender node generate higher FF values.

$$C_M = \frac{1}{R}\sqrt{(x - \overline{x})^2 + (y - \overline{y})^2} \quad (5)$$

To add QORE with the DB SRP, we establish two criteria as input to FF (Eq. (6)): vehicle positioning and QoE-indicators, allowing a cross-layer VR selection in addition to the only positioning parameters of the pure DB method. As defined in Eq. (5), the DB method does not exchange messages containing location or mobility information. Thus, DQORE also maintains the statistical and receiver-based features of the DB method when determining the best VR options. Further, QORE can be coupled with other SRPs by simply changing the parameters in the DBF phase for the VR selection process. It might be suitable for link quality-based, stochastic-based, or CB SRPs. Depending on the routing strategy, the QORE steps in the DBF phase can be easily adapted.

3.3. Persistent Multi-hop Forwarding (PMF)

As video transmissions are often long (e.g., 20 s), whenever a given node wins the DBF phase, VS transmits video packets explicitly without any additional delay and in a backbone fashion (t_2 in Figs. 2e and f). Thereby, QORE reduces additional delays and packet duplication from the DBF phase by introducing the Persistent Multi-hop Forwarding (PMF) phase. During the transmission, the video content should be delivered even in the presence of node failures or channel variations. QORE detects routing failures, providing a smoother backbone management. In particular, QORE considers that every node that composes the video dissemination backbone $P(VS, VD)$ should perceive whether it is still a reliable or valid route to transmit packets. This is achieved by receiving reply messages. We define a control packet, called Peer Quality Message (PQM), which contains the $\overline{D}$ perceived by each forwarder. Thus, if a given VR_2 receives a video flow from VR_1, it must compute $\overline{D}$ perceived in each $W(VF)$ and send a $PQM_{W(VF)}$ to VR_1.

Any built backbone returns to the DBF phase, when it detects that the video quality falls below a predefined video distortion threshold ($\overline{D}_t$). Further, any node that composes a transmitting backbone considers that the route is not valid anymore, as long as it does not receive any reply message from its previous VR within a certain period of time, i.e., timeout $= 0.5s$. Hence, it returns to the DBF phase to re-establish a new backbone.

Upon computing C_{QoE} and C_D, each VR contains the calculated criteria, i.e., $D = \{C_{QoE}, C_D\}$ ($|D| = 2$). Thus, considering the different weights $\omega_p | \sum_{p=1}^{|D|} \omega_p = 1$, Eq. (6) calculates FF by multiplying the values d_p in D and the weights of evaluation criteria, similarly to others multi-criteria approaches [Rosário et al. 2014]. From Eq. (6), other criteria can be added to FF of QORE, depending only on the underline SRP.

$$FF = \sum_{p=1}^{|D|} (d_p \times \omega_p) \tag{6}$$

4. Performance Evaluation

This section shows the methodology and metrics used to evaluate the transmitted video flows, where DQORE is compared to main related works. To establish a relevant scenario, we have considered a 5 square kilometers of West Los Angeles from OpenStreetMap, which was imported into SUMO (Simulation of Urban MObility). It allows us to reproduce the desired vehicle movements and V2V interactions according to empirical data. Further, in our simulations, vehicles move with a speed ranging from 6 to 20 m/s, where each one holds an IEEE 802.11p (5.89 GHz, 6 Mbps) radio with about 250 m transmission range and a TB_{Max} of 30 pkts. By following the approach proposed in [Torres et al. 2014], we scheduled an accident situation, so that when VS perceives the accident, it starts the dissemination of VF. Each VF must be received by vehicles at a distance lower than 2.5 km from VS, providing limitation of hops, as proposed in [Di Felice et al. 2013]. The Nakagami Fading Channel was used as propagation model.

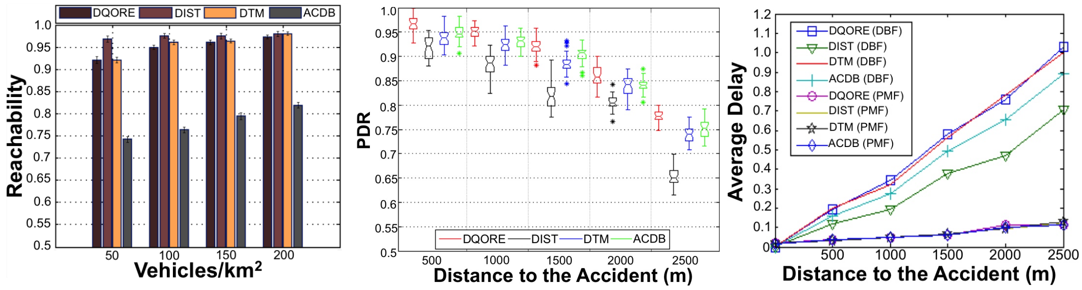
Aiming for realistic results, we have used EvalVid - A Video Quality Evaluation Tool-set that allows us evaluating the video quality. Thus, we have conducted the experiments by transmitting real MPEG-4 sequences (720 x 480 pixels) lasting approximately 25 s, available in [Video Sequences 2014], with 768 kbps and 24 fps, internal GoP structure (size 20) configured as two B-frames for each P-frame, and interleaving distance $d = 2$. The videos and the road/vehicle features were added into Network Simulator 2.33.

To demonstrate the impact of DQORE in broadcasting QoE-aware video flows in VANETs, we used a straightforward SRP built from the DB method (named DIST), DTM [Slavik et al. 2014], and ACDB [Torres et al. 2014] for comparison. The DTM and ACDB protocols use DB and CB methods, respectively. In DTM, the farther away the node to the spatial mean, the shorter the $BackoffTimer$. ACDB uses the density information provided by a beaconing approach to dynamically adjust a counter and the maximum $BackoffTimer$ before rebroadcasting packets. These protocols were adjusted with the PMF phase to reduce the contention phase: once a vehicle successfully transmits video packets, its timer for the next packets will be zero. We introduced these improvements because the standard protocols, as they were, did not represent a fair comparison.

The I-, P,- and B-frame weights (α_1, α_2, and α_3) affect the QORE performance. We have conducted independent empirical evaluations and we concluded that $\alpha_1 = 0.65$, $\alpha_2 = 0.3$, and $\alpha_3 = 0.05$ give the best C_{QoE} results. Further, the weights for each criterion ω_1 and ω_2 were fixed in 0.6 and 0.4, respectively, which allow QORE achieve the best trade-off between lowest number of hops and enough QoE-indicators to an acceptable video quality. In addition, we set CW_{Max} to 100 ms, $W(VF)$ to 80 ms, and $\overline{D}_t$ to 0.75.

The above protocols were assessed by Packet Delivery Rate (PDR), average delay, and reachability, which shows the average fraction of nodes that receive the broadcasted videos. Since measuring the human experience is key for our work, QoE-based measurements were carried out with Structural SIMilarity (SSIM) and Mean Opinion Score (MOS). SSIM is a objective QoE metric that measures the structural distortion of the video to obtain a better correlation with the user's perspective. For the subjective experiments (MOS), an Android application [Felice et al. 2014] was used (following the ITU-R rules) to playback the transmitted videos and collect their evaluations. We used the Single Stimulus method of ITU-R BT.500, with 30 subjects (ranging from 18 to 40 years old), where after watching a video, each viewer assesses the video quality level by selecting a score ranging from 1 (poor) to 10 (excellent). The distorted videos were played on a tablet (8.4-inch) placed in the back of the headrest of a car. The results are shown with variations in the number of vehicles (50 - 200 veh/km^2) and distance to the crash area (500 - 2500 m), being an average of 35 simulations (95% confidence level). Each simulation lasts 500 s, where, a VS sends a VF at any time after the initial 100 s and before the last 100 s. At receivers, the decoder uses Frame-Copy as the error concealment technique.

Regarding network performance in reachability, PDR, and average delay, Figs. 3a-c show the performance results for the four simulated protocols. From Fig. 3a, DQORE, DIST, and DTM notably outperform ACDB in terms of reachability. This is because ACDB does not uses positioning parameters to relay node selection (CB SRP). On average, DQORE and DTM increase the reachability by 18.6% and 18.9% compared to ACDB, respectively, while DIST increases reachability by 2.4% and 2.1% over DQORE and DTM, respectively. The highest reachability of DIST occurs due to the selection criteria of DQORE and DTM, where the forwarding vehicles are (potentially) close vehicles that experience more stable link connectivity, allowing a higher PDR, as shown in Fig. 3b. Thus, by Figs. 3a and b, DIST achieves a high reachability, but faces several broken link situations leading to a low PDR. Otherwise, ACDB reaches a PDR slightly higher



(a) Reachability (b) PDR (c) Average Delay

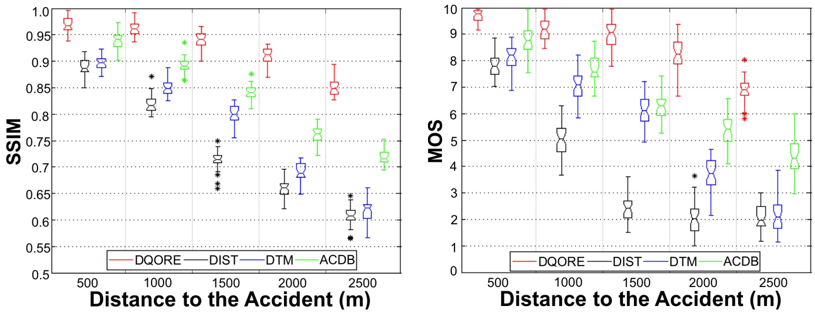
Figure 3. QoS-based metrics for all protocols with 200 veh/km^2

in comparison to DTM, i.e., around 4.3%. This is because sometimes DTM elects farthest relay nodes such as DIST, mainly when there are few neighboring vehicles. Further, ACDB adapts FF_t depending on the number of neighbors, thus, increasing PDR.

The impact of the DBF and PMF phases are significant on the delivery delay over the transmission (Fig. 3c), since PMF allows a great reduction of the average delay by using a contention-free forwarding. Here, we consider the average delay required by a $W(VF)$ to be transmitted in a given range starting from VS. DIST at DBF phase experiences the lower delay, due to the reduced number of hops achieved by the DB method. After, DIST is followed by ACDB, since this protocol reduces its $BackoffTimer$ when in presence of few nodes. Also, when DIST is compared with DQORE and DTM, the delay reduction provided by DIST is around 45.4% and 44.8%, respectively for 1000m, and 46.0% and 46.5% for 2000m of range. As mentioned previously, DQORE, according to its forwarding criteria, provides more effort to deliver flows with better quality, this could mean forwarding streams to alternative nodes, increasing transmission durations. Otherwise, DIST just tries to minimize the number of hops and does not make any effort. This might result in longer delays and path lengths for DQORE. However, the achieved delay levels are negligible even in video applications and are significantly lower than the requirements of 4 to 5 s defined by CISCO [Hatting et al. 2005].

As discussed before, QoS-based metrics (e.g., PDR) are not enough to measure the quality level from the user's perspective. Thus, aiming to understand and confirm the impact of the QoE-Indicators criterion, the results in Fig. 4 present SSIM and MOS. SSIM values range from 0 to 1, where a higher value means better video quality. In Fig. 4.a, DQORE keeps the SSIM values around 0.97 and 0.85. An average increase of 23.4%, 16.3%, and 10.8% compared to DIST, DTM, and ACDB, respectively. It presents more deeply results than those obtained in Fig. 3b and shows significant benefits to the user's experience. This occurred because DQORE can estimates when the quality of the transmitting flow decreases based on the different received frame types, codec configurations, and losses, allowing vehicles, through FF calculation, to switch to others nodes, before increasing damage on the flow quality. For instance, let's suppose a $W(VF_i)$ successfully received by a VR_i in $|W(VF_i)|$ ms. As the spatial distribution of vehicles does not change very quickly in a short period of time (e.g., 3 s), it is likely that VR_i, continue to receive successfully a greater number of packets until a new route becomes necessary. Thus, DQORE provides a trade-off between hop-length and video quality.

With respect to real-time video assessment, sometimes the correlation between the SSIM results and subjective scores does not have a high accuracy. In face of this, it



(a) Metric: SSIM; GoP size: 14 (b) Metric: MOS; GoP size: 20

Figure 4. QoE-based metrics for all protocols with 200 veh/km^2

is fundamental to have MOS experiments in order to really understand the video quality level according to human perception [Aguiar et al. 2014]. The results in the Fig. 4.b reinforce the results in the Fig. 4.a. Besides that, these results demonstrate that the QoE-indicators metric can be successfully extended from the distortion model introduced by Shu Tao in [Tao et al. 2008] and performs well when employed in QORE. Fig. 4.b presents the average MOS scores for all protocols and confirms that DQORE allows the delivery of live video sequences with a good or excellent quality in urban VANETs.

5. Conclusion

This paper introduced QORE to real-time video dissemination with QoE-awareness in VANETs. QORE aims to share videos with a better quality than existing works, since it employs the Interleaving scheme and QoE-indicators criterion to support the selection of the best next hops, changing to other nodes as soon as lower quality is identified. Simulation results highlight the performance and QoE-awareness support of QORE by measuring the video quality levels when the distance to the sender varies. By creating backbones and according to its forwarding criteria, QORE provides a greater support to a self-organized dissemination of video flows with a higher quality from the user's point-of-view. This could mean forwarding of streams to alternative nodes by increasing transmission durations, but nonetheless, still are insignificant to real-time video requirements. In future works, we will perform a study with different video features (e.g., varying the GoP size and packetization), so that FF_t can be dynamically adjusted to allow better performance.

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